

## Train Your Network Report

### Initial Network

Model: "sequential"

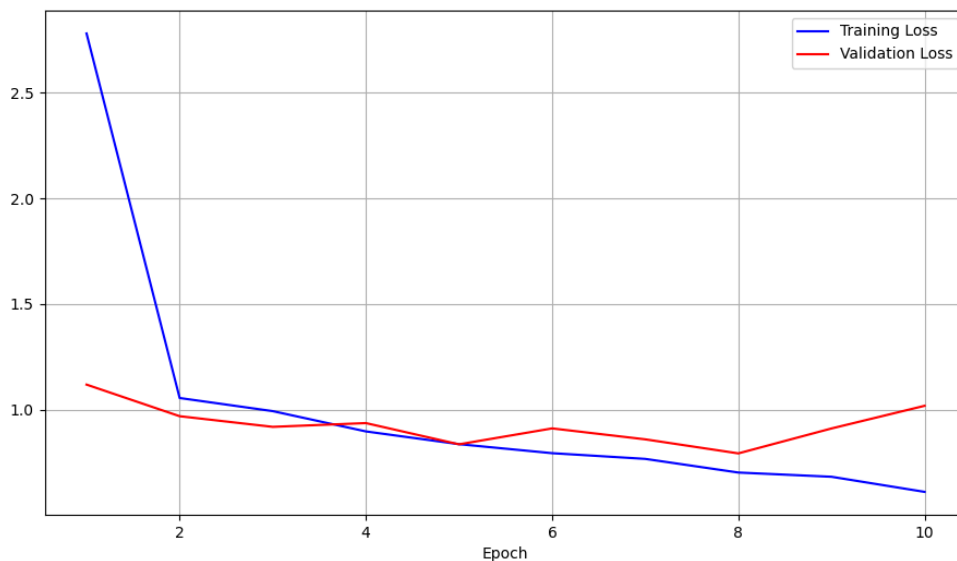
| Layer (type)                    | Output Shape         | Param # |
|---------------------------------|----------------------|---------|
| rescaling (Rescaling)           | (None, 150, 150, 3)  | 0       |
| conv2d (Conv2D)                 | (None, 148, 148, 16) | 448     |
| max_pooling2d (MaxPooling2D)    | (None, 74, 74, 16)   | 0       |
| conv2d_1 (Conv2D)               | (None, 72, 72, 32)   | 4640    |
| max_pooling2d_1 (MaxPooling 2D) | (None, 36, 36, 32)   | 0       |
| flatten (Flatten)               | (None, 41472)        | 0       |
| dense (Dense)                   | (None, 64)           | 2654272 |
| dense_1 (Dense)                 | (None, 3)            | 195     |

Total params: 2,659,555

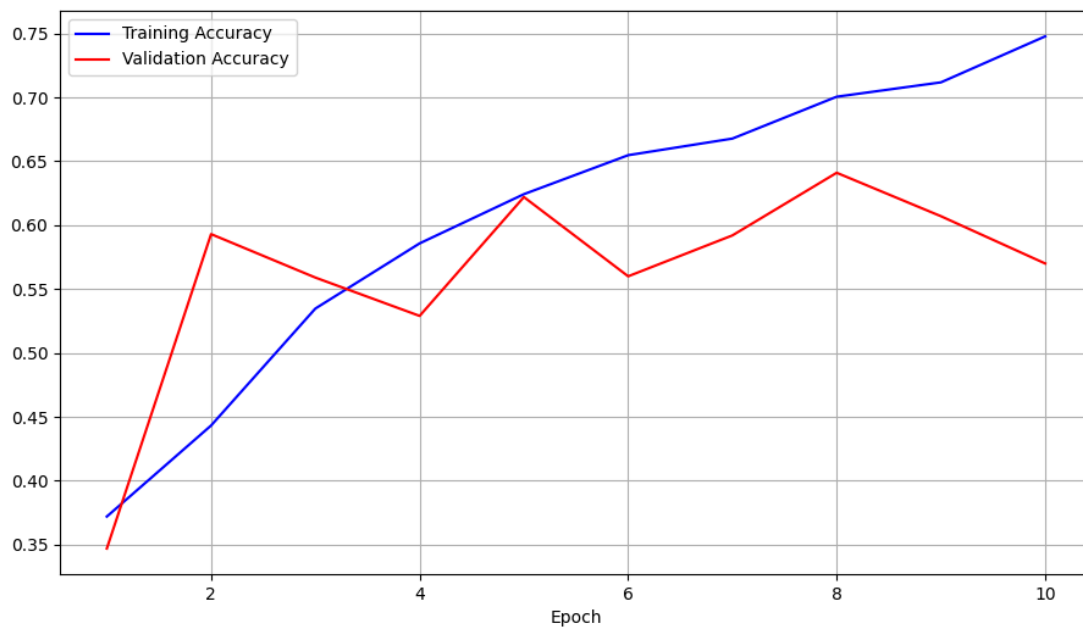
Trainable params: 2,659,555

Non-trainable params: 0

### Plot Showing Training and Validation Loss as Function of Epoch



**Plot Showing Accuracy Against the Training and Validation Sets as a Function of Epoch:**



**Accuracy and Loss of Best Learned model:**

Epoch 10/10

32/32 [=====] - 16s 505ms/step - loss: 0.6125 - accuracy: 0.7477 -  
val\_loss: 1.0193 - val\_accuracy: 0.5700

Test Set Evaluation:

\* Evaluating basic\_model

30/30 [=====] - 4s 128ms/step - loss: 0.9314 - accuracy: 0.6133